



Semester 2 Examinations 2012/2013

Exam Codes	3BME1, 4BME1, 3BMS2, 3BS3, 3BS9, 4BS4, 3CS2, 4CS2 1AH2, 1PAH1, 1HA1, 1HDS1
Exams	Arts and Science Examinations
Module	Numerical Analysis II
Module Codes	MA378, MA531
External Examiner	Dr Colin Campbell
Internal Examiner(s)	Prof. Graham Ellis Dr Emil Sköldbberg
<u>Instructions:</u>	Answer <i>three</i> questions.
Duration	Two Hours
No. of Pages	4 pages, including this one
School	School of Mathematics, Statistics and Applied Mathematics
<u>Requirements:</u>	No special requirements
Release to Library:	Yes
Statistical Tables/ Log Tables	Yes

Q1. (a) Given $n + 1$ distinct points, $x_0 = 0, x_1 = 1, \dots, x_n = n$; define

$$L_i(x) = \prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j},$$

and show that

$$L_1(x) + 2L_2(x) + 3L_3(x) + \dots + nL_n(x) = x$$

(b) Suppose f is a $2n + 2$ times differentiable real-valued function on the interval $[a, b]$. Suppose further that $n \geq 0$ and that $x_0 < x_1 < \dots < x_n$ are $n + 1$ distinct points in $[a, b]$.

(i) Show that the polynomial

$$p(x) = \sum_{i=0}^n f(x_i)H_i(x) + \sum_{i=0}^n f'(x_i)K_i(x)$$

satisfies

$$p(x_i) = f(x_i), \quad p'(x_i) = f'(x_i), \quad i = 0, \dots, n,$$

where

$$H_i(x) = [L_i(x)]^2(1 - 2L'_i(x_i)(x - x_i)), \quad K_i(x) = [L_i(x)]^2(x - x_i).$$

Furthermore, show that $p(x)$ is the *only* polynomial of degree at most $2n + 1$ satisfying these conditions.

(ii) Show that for each $x \in [a, b]$ there is a $\xi \in (a, b)$ such that

$$f(x) - p(x) = \frac{f^{(2n+2)}(\xi)}{(2n+2)!} [\pi_{n+1}(x)]^2$$

where $\pi_{n+1}(x) = \prod_{i=0}^n (x - x_i)$.

- Q2.** (a) Define the *piecewise linear spline* interpolating a function $f(x)$ at the points $a = x_0 < x_1 < \cdots < x_n = b$.
- (b) Suppose that $l(x)$ is the piecewise linear spline as in (a) and that the points are evenly spaced, with $h = x_i - x_{i-1}$. Use Cauchy's theorem to give an upper bound for $\|f - l\|_\infty$ in terms of h and $\|f''\|_\infty$.
- (c) Let $f(x) = \arctan(x)$, $a = 0$ and $b = 1$. What value of n would you need to ensure that $\|f - l_n\|_\infty < 5 \cdot 10^{-5}$?
- (d) Let $f \in C^1[a, b]$, and let $a = x_0 < x_1 < \cdots < x_n = b$ for $m \geq 2$. Define the Hermite cubic spline $s(x)$, interpolating $f(x)$ at the knot points $x_0, \dots, x_n$, which are assumed to be evenly spaced with $h = x_i - x_{i-1}$.

Show that if the Hermite cubic spline is given on the interval $[x_{i-1}, x_i]$ by

$$s(x) = c_0 + c_1(x - x_{i-1}) + c_2(x - x_{i-1})^2 + c_3(x - x_{i-1})^3,$$

then

$$\begin{aligned} c_0 &= f(x_{i-1}), & c_2 &= 3 \frac{f(x_i) - f(x_{i-1})}{h^2} - \frac{f'(x_i) + 2f'(x_{i-1})}{h}, \\ c_1 &= f'(x_{i-1}), & c_3 &= \frac{f'(x_i) + f'(x_{i-1})}{h^2} - 2 \frac{f(x_i) - f(x_{i-1})}{h^3}. \end{aligned}$$

- Q3.** (a) Explain Newton-Cotes quadrature, and show how to derive the 2-point and 3-point Newton-Cotes quadrature rules (i.e. the trapezium method and Simpson's method) for approximating the integral

$$\int_{-1}^1 f(x) dx.$$

- (b) Use the composite Trapezoid rule with 4 subintervals to approximate

$$\int_0^1 e^{-x^2} dx.$$

Give a bound for the error of your approximation.

- (c) Let $\tilde{p}_0(x), \tilde{p}_1(x), \dots$ be the sequence of polynomials characterised by (i) $\tilde{p}_i(x)$ is a monic polynomial of degree i for all i and (ii), $\tilde{p}_i(x)$ and $\tilde{p}_j(x)$ are orthogonal with respect to the inner product $(u, v) = \int_a^b u(x)v(x) dx$ for all i, j .

Let $x_0, \dots, x_n$ be the zeros of $\tilde{p}_{n+1}(x)$ and define the n -point Gaussian quadrature rule G_n as

$$G_n(f) = \sum_{i=0}^n w_i f(x_i), \quad w_i = \int_a^b L_i(x)^2 dx,$$

where the $L_i(x)$ are the Lagrange polynomials. Show that G_n has precision $2n + 1$.

- Q4.** (a) Consider the following boundary value problem

$$-u''(x) + u(x) = x, \quad u(0) = u(3) = 0.$$

Use the finite element method with piecewise linear basis functions on the mesh $\{0, 1, 2, 3\}$ to find an approximation of the solution to this problem.

- (b) State and prove Céa's lemma, and explain its relevance to the errors in the finite element method.